

```
Last login: Mon Jun 22 12:46:16 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % cd /Users/adityadave/PE/PE-Daywise-Work-Tracker/19-JUNE/Q2\&Q21
adityadave@Adityas-MacBook-Air-3 Q2&Q21 % python3 script.py
Running 1/10...
Running 2/10...
Running 3/10...
Running 4/10...
Running 5/10...
Running 6/10...
Running 7/10...
Running 8/10...
Running 9/10...
Running 10/10...
```

10TH RUN OUTPUT

```
SET
WARNING: Log 1: Planner Input:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname part
  :colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
    "p_container" "p_retailprice" "p_comment")
  }
  :rtekind 0
  :relid 17169
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  {RANGETBENTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname supplier
  :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
    s_acctbal" "s_comment")
  }
  :rtekind 0
  :relid 17181
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals (
    {BOOLEXP
    :bbool and
    :args (
      {OPEXPR
      :opno 1756
      :opfuncid 1720
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
        :varno 2
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 6
        :location -1
        }
        {FUNCEXP
        :funcid 1740
        :funcresulttype 1700
        :funcresultset false
        :funcvariadic false
        :funcformat 2
        :funccollid 0
        :inputcollid 0
        :args (
          {CONST
          :consttype 23
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location -1
          :constvalue 4 { 0 0 0 0 0 0 0 }
          }
        )
        :location -1
        }
      )
    }
  )
  :location -1
  }
  {SUBLINK
  :subLinkType 0
  :subLinkId 0
  :testexpr <>
  :operName <>
  :subselect
  {QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSublinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :ctelist <>
```

supplier.s_acctbal > 0

```

:rtable (
  {RANGETBLENTY
  :alias
    {ALIAS
      :aliasname n
      :colnames <>
    }
  :eref
    {ALIAS
      :aliasname n
      :colnames ("n_nationkey" "n_name" "n_regionkey" "n_commen
t")
    }
  :rtekind 0
  :relid 17158
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  {RANGETBLENTY
  :alias
    {ALIAS
      :aliasname r
      :colnames <>
    }
  :eref
    {ALIAS
      :aliasname r
      :colnames ("r_regionkey" "r_name" "r_comment")
    }
  :rtekind 0
  :relid 17164
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 17158
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 10)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :relid 17164
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
  )
  }
  :quals
    {BOOLEXP
    :boolop and
    :args (
      {OPEXP
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
        }
        {VAR
        :varno 2
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 4
        :location -1
        }
      )
    }
    :location -1
  )
  }
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
  }
  )
  }
  }

```

n.n_nationkey =
supplier.s_nationkey

n.n_regionkey =
r.r_regionkey

```

        :varnosyn 2
        :varattnosyn 1
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 2
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
    }
    {CONST
        :consttype 1042
        :constypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :constypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 17191
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
    {BOOLEXP
        :boolop and
        :args (
            {OPEXPR
                :opno 521
                :opfuncid 147
                :opresulttype 16
                :opretset false
                :opcollid 0
                :inputcollid 0
            }
            {VAR
                :varno 3
                :varattno 3
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 3
                :varattnosyn 3
                :location -1
            }
            {CONST
                :consttype 23
                :constypmod -1
                :constcollid 0
                :constlen 4
                :constbyval true
                :constisnull false
                :location -1
            }
        )
    }
)
}

```

r.r_name = 'EUROPE'

ps_availqty > 100

```

:constvalue 4 [ 100 0 0 0 0 0 0 0 ]
}
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey"
"s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 17181
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varatype 1700
:varitypm 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
}
:location -1
}
)
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionke
y" "n_comment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>

```

s.s_acctbal > 0

```

}
{RANGETBLENTY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment"
)
}
:rtekind 0
:relid 17164
:inh true
:rtekind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varrtno 1
:varstype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varrtnosyn 1
:location -1
}
{VAR
:varno 1
:varrtno 4
:varstype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varrtnosyn 4
:location -1
}
}
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varrtno 3
:varstype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varrtnosyn 3
:location -1
}
{VAR
:varno 2
:varrtno 1
:varstype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varrtnosyn 1
:location -1
}
}
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varrtno 2
:varstype 1042
:varitypmode 29
:varcollid 100
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen 3
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 8
0 69 ]
}
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
}
:location -1
}
}
{OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6

```

```

:varstype 1700
:vartypemod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypemod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
):location -1
}
):location -1
}
):location -1
}
):location -1
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypemod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
):location -1
}
):location -1
}
):location -1
}
}
{RANGEBTENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtkindex 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGEBTENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtkindex 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rtperminfos (
{RTEPERMISSIONINFO
:relid 17169
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
)

```

```

{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17188
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
{VAR
:varno 3
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 295
}
)
:location 293
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 318
}
{VAR
:varno 3
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 330
}
)
:location 328
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 353
}
{CONST
:consttype 23

```



```

        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 362
        :constvalue 4 [ 15 0 0 0 0 0 0 0 ]
    }
}
:location 360
}
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 1
:varattno 5
:vartype 1043
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 377
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 389
:constvalue 10 [ 40 0 0 0 0 37 66 82 65 83 83 ]
}
}
:location 384
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 410
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 424
}
}
:location 422
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 448
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 462
}
}
:location 460
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 5
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 493
}
}

```

```

{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 509
:constvalue 10 [ 40 0 0 0 0 69 85 82 79 80 69 ]
}
}
:location 507
}
{OPEXPR
:opno 1752
:opfuncid 1718
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 4
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location 530
}
{SUBLINK
:sublinktype 4
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
}
:rtekind 0
:relid i7191
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{BOOLEXP
:boolop and
:args (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 0 ]
}
)
}
:location -1
}
{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s

```

```

:colnames ("s_supkey" "s_name" "s_address"
" "s_nationkey" "s_phone" "s_acctbal" "s_c
omment")
}
:rtekind 0
:relid 17181
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
}
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_na
me" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_na
me" "r_comment")
}
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true

```

```

:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0
0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
}
}
)

```



```

    }
  }
  :mergeActionList <>
  :mergeTargetRelation 0
  :mergeJoinCondition <>
  :targetList (
    {TARGETENTRY
      :expr
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
      }
      :resno 1
      :resname ?column?
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDecls <>
  :withCheckOptions <>
  :stmt_location -1
  :stmt_len -1
}
:location -1
}
}
:location -1
}
}
}
{RANGETBLENTY
  :alias <>
  :eref
  {ALIAS
    :aliasname supplier
    :colnames ("s_suppkkey" "s_name" "s_address" "s_nationk
ey" "s_phone" "s_acctbal" "s_comment")
  }
  :rtekind 0
  :relid 17181
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals (
    {BOOLEXP
      :boolop and
      :args (
        {OPEXP
          :opno 1756
          :opfuncid 1720
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            {VAR
              :varno 2
              :varattno 6
              :vartype 1700
              :vartypmod 983046
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 2
              :varattnosyn 6
              :location -1
            }
            {FUNCEXP
              :funcid 1740
              :funcresulttype 1700
              :funcresultset false
              :funcvariadic false
              :funcformat 2
              :funccollid 0
              :inputcollid 0
              :args (
                {CONST
                  :consttype 23
                  :consttypmod -1
                  :constcollid 0
                  :constlen 4
                  :constbyval true
                  :constisnull false
                  :location -1
                  :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
                }
              )
            }
          )
          :location -1
        }
      )
    }
  )
  :location -1
}
}
{SUBLINK
  :subLinkType 0
  :subLinkId 0
  :testexpr <>
  :operName <>
  :subselect
    {QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStmt <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSubLinks false
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
      :hasForUpdate false
      :hasRowSecurity false
      :hasGroupRTE false
      :isReturn false
      :ctelist <>
      :rtable (
        {RANGETBLENTY

```

```

:alias
{
  ALIAS
  :aliasname n
  :colnames <>
}
:eref
{
  ALIAS
  :aliasname n
  :colnames ("n_nationkey" "n_name" "n_regio
nkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
(RANGETBLENTTRY
:alias
{
  ALIAS
  :aliasname r
  :colnames <>
}
:eref
{
  ALIAS
  :aliasname r
  :colnames ("r_regionkey" "r_name" "r_comme
nt")
}
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointrree
{FROMEXPR
:fromlist {
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
}
}
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
}
}
}
}

```

```

        :varattnosyn 1
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 2
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
    }
    {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 7
            9 80 69 ]
    }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_com
ment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
}
rteperminfos (
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181

```



```

:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
{fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 812
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 824
}
}
:location 822
}
)
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 863
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 875
}
}
:location 873
}
)
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 914
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)

```

```

        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location 928
    }
}
:location 926
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 3
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 3
        :location 968
    }
    {VAR
        :varno 4
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 1
        :location 982
    }
}
:location 980
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 4
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location 1029
    }
    {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location 1045
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
}
:location 1043
}
)
:location 859
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {AGGREF
            :aggfnoid 2146
            :aggtype 1700
            :aggcollid 0
            :inputcollid 0
            :aggtranstype 0
            :aggargtypes (o 1700)
            :aggdirectargs <>
            :args (
                {TARGETENTRY
                    :expr
                    {VAR
                        :varno 1
                        :varattno 4
                        :vartype 1700
                        :vartypmod 983046
                        :varcollid 0
                        :varnullingrels (b)
                        :varlevelsup 0
                        :varreturningtype 0
                        :varnosyn 1
                        :varattnosyn 4
                        :location 599
                    }
                    :resno 1
                    :resname <>
                    :ressortgroupref 0
                    :resorigtbl 0
                    :resorigcol 0
                    :resjunk false
                }
            )
        }
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar false
        :aggvariadic false
        :aggkind n
        :aggpresorted false
        :agglevelsup 0
        :aggsplit 0
        :aggno -1
        :aggtransno -1
        :location 595
    }
    :resno 1
    :resname min
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>

```

and region.r_name =
'EUROPE'

```

        :groupClause <>
        :groupDistinct false
        :groupingSets <>
        :havingQual <>
        :windowClause <>
        :distinctClause <>
        :sortClause <>
        :limitOffset <>
        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location 0
        :stmt_len 0
    }
    :location 546
}
}
:location 544
}
}
:location 314
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location 51
}
:resno 1
:resname s_acctbal
:ressortgroupref 1
:resorigtbl 17181
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 70
}
:resno 2
:resname s_name
:ressortgroupref 3
:resorigtbl 17181
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 86
}
:resno 3
:resname n_name
:ressortgroupref 2
:resorigtbl 17158
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 17169
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 17169
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:varatype 1042
:varitypmid 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:varatype 1043
:varitypmid 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1752
:sortop 1756
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 3
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 4
:eqop 96
:sortop 97
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount
{FUNCEXP
:funcid 481
:funcresulttype 20
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1153
:constvalue 4 [ 100 0 0 0 0 0 0 ]
}
)
:location -1
}
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=481 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0

```

Log 1 Done there are so many quals because in partsupp policy it is using supplier and supplier also has policy so we have nested conditions (security Qals).

```

:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname n
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
      }
    :rtekind 0
    :relid 17158
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname r
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
      }
    :rtekind 0
    :relid 17164
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17158
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17164
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
  )
}
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
      {PARAM
        :paramkind 1
        :paramid 2
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
    :location -1
  }
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
      }
      {VAR
        :varno 2
        :varattno 1

```

```

:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
}
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:varstype 1042
:varstypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
}{CONST
:consttype 1042
:conststypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
}
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :varstype 23 :varstypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramstypmod -1 :paramcollid 0 :location -1} :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :varstype 23 :varstypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramstypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=2 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :varstype 23 :varstypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramstypmod -1 :paramcollid 0 :location -1} :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :varstype 23 :varstypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramstypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 2 :varattno 2 :varstype 1042 :varstypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 1042 :conststypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :varstype 1042 :varstypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :conststypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :varattno 1 :varstype 23 :varstypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 1 :varstype 23 :varstypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {
QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:targetList <>
:rttable (
{RANGELEFTENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}

```

```

    }
    :rtekind 0
    :relid 17158
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTY
  :alias
  {ALIAS
  :aliasname r
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname r
  :colnames ("r_regionkey" "r_name" "r_comment")
  }
  :rtekind 0
  :relid 17164
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 17158
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 10)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :relid 17164
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9)
  :insertedCols (b)
  :updatedCols (b)
  }
)
):jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    {RANGETBLREF
    :rtindex 2
    }
  )
  :quals (
    {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
      }
      {VAR
      :varno 2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
      }
    )
    :location -1
  )
  {OEXPR
  :opno 1054
  :opfuncid 1048
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 100
  :args (
    {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
    }
    {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
  )
  :location -1
}
)
):mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 1

```

```
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 2 :varattno 2 :vartype
1042 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :co
nstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :con
stvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}}
WARNING: is_parallel_safe: OUTPUT qual = {(RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 2 :varattno 2 :vartype
1042 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :c
onstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :con
stvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}} | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)
}
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 3 :vartype 23 :va
rtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :
has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jd
omain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}}
WARNING: is_parallel_safe: OUTPUT qual = {(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :
has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}} | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
}
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
}
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupTE false
:isReturn false
:cteList <>
:table (
:table (
:RANGELENTTRY
:alias
:alias
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17181
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
(OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
```



```

:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {CONST
    :consttype 1700
    :consttypmod -1
    :constcollid 0
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 6 [ 24 0 0 0 0 -128 ]
  }
)
:location -1
}
{ALTERNATIVESUBPLAN
:subplans (
  {SUBPLAN
    :subLinkType 0
    :testexpr <>
    :paramids <>
    :plan_id 1
    :plan_name SubPlan\ 1
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse true
    :parallel_safe false
    :setParam <>
    :parParam (i 2)
    :args (
      {VAR
        :varno 1
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 4
        :location -1
      }
    )
    :startup_cost 0
    :per_call_cost 24.2625
  }
  {SUBPLAN
    :subLinkType 2
    :testexpr
    {OEXPR
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 4
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 4
          :location -1
        }
        {PARAM
          :paramkind 1
          :paramid 3
          :paramtype 23
          :paramtypmod -1
          :paramcollid 0
          :location -1
        }
      )
    }
    :location -1
  }
  :paramids (i 3)
  :plan_id 2
  :plan_name SubPlan\ 2
  :firstColType 23
  :firstColTypmod -1
  :firstColCollation 0
  :useHashTable true
  :unknownEqFalse true
  :parallel_safe true
  :setParam <>
  :parParam <>
  :args <>
  :startup_cost 24.4875
  :per_call_cost 0.0025
}
)
})
}
)
:rteperinfos (
  {RTEPERMISSIONINFO
    :relid 17181
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
:quale (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23

```

```

:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paritypmod -1
:paramcollid 0
:location -1
}
}
:location -1
}
{OPEXPR
:opno 1756
:opfncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 6
:varityp 1700
:varitypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
){CONST
:consttype 1700
:constitypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningNewAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

```

WARNING: is_parallel_safe: INPUT qual = {(RESTRICTINFO :clause {OPEXPR :opno 1756 :opfncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 6 :varityp 1700 :varitypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1700 :constitypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :outer_select -1 :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0} (RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans {(SUBPLAN :sublinktype 0 :testexpr <> :paramids <> :plan_id 1 :plan_name SubPlan 1 :firstColType 2278 :firstColTypeMod -1 :firstColCollation 0 :useHashTable false :unknownEqfalse true :parallel_safe false :setParam <> :parParam (i 2) :args {(VAR :varno 1 :varattno 4 :varityp 23 :varitypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) :startup_cost 0 :per_call_cost 24.2625} (SUBPLAN :sublinktype 2 :testexpr {OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 4 :varityp 23 :varitypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paritypmod -1 :paramcollid 0 :location -1} :location -1} :plan_id 2 :plan_name SubPlan 2 :firstColType 23 :firstColTypeMod -1 :firstColCollation 0 :useHashTable true :unknownEqfalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0} (RESTRICTINFO :clause {OPEXPR :opno 1756 :opfncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 6 :varityp 1700 :varitypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1700 :constitypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0} (RESTRICTINFO :clause {OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 1 :varityp 23 :varitypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paritypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0} (RESTRICTINFO :clause {OPEXPR :opno 1756 :opfncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 1 :varityp 23 :varitypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :pa rankid 1 :paramid 1 :paramtype 23 :paritypmod -1 :paramcollid 0 :location -1} :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0}
WARNING: max_parallel_hazard_walker: SubPlan plan_id=1 parallel_safe=0 reason=correlated_subplan

```

```

WARNING: is_parallel_safe: OUTPUT qual = {(RESTRICTINFO :clause {OPEXPR :opno 1756 :opfncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 6 :varityp 1700 :varitypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1700 :constitypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0} (RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans {(SUBPLAN :sublinktype 0 :testexpr <> :paramids <> :plan_id 1 :plan_name SubPlan 1 :firstColType 2278 :firstColTypeMod -1 :firstColCollation 0 :useHashTable false :unknownEqfalse true :parallel_safe false :setParam <> :parParam (i 2) :args {(VAR :varno 1 :varattno 4 :varityp 23 :varitypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) :startup_cost 0 :per_call_cost 24.2625} (SUBPLAN :sublinktype 2 :testexpr {OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 4 :varityp 23 :varitypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paritypmod -1 :paramcollid 0 :location -1} :location -1} :plan_id 2 :plan_name SubPlan 2 :firstColType 23 :firstColTypeMod -1 :firstColCollation 0 :useHashTable true :unknownEqfalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0} (RESTRICTINFO :clause {OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 1 :varityp 23 :varitypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :pa rankid 1 :paramid 1 :paramtype 23 :paritypmod -1 :paramcollid 0 :location -1} :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0} | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
  QUERY:
    commandType 1
    querySource 0
    canSetTag true
    utilityStmt <>
    resultRelation 0

```

```

:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname n
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
      }
    :rtekind 0
    :relid 17158
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname r
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
      }
    :rtekind 0
    :relid 17164
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17158
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17164
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
  )
}
:quals (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
      {PARAM
        :paramkind 1
        :paramid 5
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
    :location -1
  }
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
      }
      {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
)
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=5 in safe_list=0 + reason=outer reference

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPPARALLEL_RESTRICTED

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1})

WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE

WARNING: **Log 2: After pull up At starting of Path Generation:**
DETAIL:

```

{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:cteList <>
:table {
{RANGETABLEENTRY
:alias
{ALIAS
:aliasname n
:colnames <n
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:reloid 17158

```

```

:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1040
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0

```

```

:varnullings (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}

:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}

:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDns <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1854 :opfuncid 1848 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 180 :args ({VAR :varno 2 :varattno 2 :vartype
1842 :vartypmod 29 :varcollid 180 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1842 :consttypmod -1 :constcollid 180 :constlen -1 :con
stbval false :constisnull false :location -1 :constvalue 10 [ 4 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :left_relds (b 2) :right_relds (b 2) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1842 :
vartypmod 29 :varcollid 180 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1842 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}) :right_em (EQUIVALENCEMEMBER :em_expr {CONST :consttype 1842 :consttypmod -1 :constcollid 180 :constlen -1 :constbval false :constisnull false :location -1 :cons
tvalue 10 [ 4 0 0 0 69 85 82 79 80 69 ]} :em_relds (b 2) :em_is_const true :em_is_child false :em_datatype 1842 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}) :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1854 :opfuncid 1848 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 180 :args ({VAR :varno 2 :varattno 2 :vartype
1842 :vartypmod 29 :varcollid 180 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1842 :consttypmod -1 :constcollid 180 :constlen -1 :con
stbval false :constisnull false :location -1 :constvalue 10 [ 4 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :left_relds (b 2) :right_relds (b 2) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1842 :
vartypmod 29 :varcollid 180 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1842 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}) :right_em (EQUIVALENCEMEMBER :em_expr {CONST :consttype 1842 :consttypmod -1 :constcollid 180 :constlen -1 :constbval false :constisnull false :location -1 :con
stvalue 10 [ 4 0 0 0 69 85 82 79 80 69 ]} :em_relds (b 2) :em_is_const true :em_is_child false :em_datatype 1842 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}) :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
{VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1}
{VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
{VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
{VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1}
{VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
{VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jd
omain {JOINDOMAIN :jd_relds (b 1 2)}) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}) :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}) :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right
```

```

:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
}
:location -1
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramids <>
:plan_id 4
:plan_name SubPlan\ 4
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 5)
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:startup_cost 0
:per_call_cost 24.2625
}
{SUBPLAN
:sublinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
{PARAM
:paramkind 1
:paramid 6
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 6)
:plan_id 5
:plan_name SubPlan\ 5
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 24.4875
:per_call_cost 0.0025
}
)
}
))
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition 0
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:smt_location -1
:smt_len -1
}

```

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 17
00 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :cons
tbval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fals
e :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seria
l 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mvfreq -1 :right_mvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {SUBPLAN :sublinktype 0 :testexpr <> :paramids <> :plan_id 4
:plan_name SubPlan 4 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 5) :args ({VAR :varno 1 :varattno
4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625} {SUBPLAN :sublinkType 2
:testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevel
sup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 6 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 6) :plan_id 5 :p
lan_name SubPlan 5 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :per_call
cost 0.0025})) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :requi
red_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :
mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvfreq -1 :right_mvfreq -1 :left_hasheqoperator 0 :right_hasheqopera
tor 0} {RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :
varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbval false :constisnull false :lo
cation -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1
:num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost
.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvfreq -1 :right_mvfreq
-1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: SubPlan plan_id=4 parallel_safe=0 → reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1
700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :cons
tbval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fals
e :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seria
l 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mvfreq -1 :right_mvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {SUBPLAN :sublinktype 0 :testexpr <> :paramids <> :plan_id 4
:plan_name SubPlan 4 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 5) :args ({VAR :varno 1 :varattno
4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625} {SUBPLAN :sublinkType 2
:testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varleve
lsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 6 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 6) :plan_id 5 :p
lan_name SubPlan 5 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :per_cal
l_cost 0.0025})) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :requi
red_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :
mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvfreq -1 :right_mvfreq -1 :left_hasheqoperator 0 :right_hasheqopera
tor 0} {RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :
varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbval false :constisnull false :lo
cation -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level
1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_co
st.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvfreq -1 :right_mvfreq
-1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:table (
{RANGEABLEENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtkind 0

```



```

:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rindex 1
}
{RANGETBLREF
:rindex 2
}
)
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 7
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1040
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)

```

```
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 7 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 7 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=7 in_safe_list=0 > reason=outer_reference

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 7 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 7 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 1054 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtekind 0
:reldid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:reldid 17164
:inh false
:relkind r
:rellockmode 1
```

```

:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
):quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
):mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>

```

```
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1054 :opfncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :co
nstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :cons
tvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1054 :opfncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :co
nstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :cons
tvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)
})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)}) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :
has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jd
omain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)}) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :
has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :var
nosyn 2 :varattnosyn 1 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
:QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:isCteList <>
:rttable (
:RANGEBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtkind 0
:reldid 17191
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:infFromCl true
:securityQuals ((
(OEXPR
:opno 521
:opfncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
(VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
)
(CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 ]
)
):location -1
)
(ALTERNATIVESUBPLAN
:subplans (
(SUBPLAN
:sublinktype 0

```

```

: testexpr <>
: paramids <>
: plan_id 3
: plan_name SubPlan\ 3
: firstColType 2278
: firstColTypmod -1
: firstColCollation 0
: useHashTable false
: unknownEqFalse true
: parallel_safe false
: setParam <>
: parParam (i 1)
: args (
  {VAR
    : varno 1
    : varattno 2
    : vartype 23
    : vartypmod -1
    : varcollid 0
    : varnullingrels (b)
    : varlevelsup 0
    : varreturningtype 0
    : varnosyn 1
    : varattnosyn 2
    : location -1
  }
)
: startup_cost 0
: per_call_cost 121512.00000000003
}
{SUBPLAN
: subLinkType 2
: testexpr
{OPEXPR
: opno 96
: opfuncid 65
: opresulttype 16
: opretset false
: opcollid 0
: inputcollid 0
: args (
  {VAR
    : varno 1
    : varattno 2
    : vartype 23
    : vartypmod -1
    : varcollid 0
    : varnullingrels (b)
    : varlevelsup 0
    : varreturningtype 0
    : varnosyn 1
    : varattnosyn 2
    : location -1
  }
  {PARAM
    : paramkind 1
    : paramid 4
    : paramtype 23
    : paramtypmod -1
    : paramcollid 0
    : location -1
  }
)
: location -1
}
: paramids (i 4)
: plan_id 6
: plan_name SubPlan\ 6
: firstColType 23
: firstColTypmod -1
: firstColCollation 0
: useHashTable true
: unknownEqFalse true
: parallel_safe false
: setParam <>
: parParam <>
: args <>
: startup_cost 121502.62500000001
: per_call_cost 0.0025
}
}
))
}
{RANGETBLENTY
: alias <>
: eref
{ALIAS
: aliasname supplier
: colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
: rtekind 0
: relid 17181
: inh false
: relkind r
: rellockmode 1
: perminfoindex 2
: tablesample <>
: lateral false
: inFromCl true
: securityQuals (
{OPEXPR
: opno 1756
: opfuncid 1720
: opresulttype 16
: opretset false
: opcollid 0
: inputcollid 0
: args (
  {VAR
    : varno 2
    : varattno 6
    : vartype 1700
    : vartypmod 983046
    : varcollid 0
    : varnullingrels (b)
    : varlevelsup 0
    : varreturningtype 0
    : varnosyn 2
    : varattnosyn 6
    : location -1
  }
  {CONST
    : consttype 1700
    : consttypmod -1
    : constcollid 0
    : constlen -1
    : constbyval false
    : constisnull false
    : location -1
    : constvalue 6 [ 24 0 0 0 0 -128 ]
  }
)
: location -1
}
}
{ALTERNATIVESUBPLAN
: subplans (
{SUBPLAN
: subLinkType 0
: testexpr <>
: paramids <>
: plan_id 7
: plan_name SubPlan\ 7
: firstColType 2278
: firstColTypmod -1
: firstColCollation 0
: useHashTable false

```

```

:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 7)
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location -1
  }
)
:startup_cost 0
:per_call_cost 24.2625
}
{SUBPLAN
:sublinkType 2
:testexpr
{OEXPR
:opno 96
:opfncid 65
:opresulttype 16
:opreset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 8
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramIds (i 8)
:plan_id 8
:plan_name SubPlan\ 8
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 24.4875
:per_call_cost 0.0025
}
}
})
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
}
)

```

```

      :updatedCols (b)
    }
  )
}
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
    {RANGETBLREF
      :rtindex 3
    }
    {RANGETBLREF
      :rtindex 4
    }
  )
}
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {PARAM
        :paramkind 1
        :paramid 0
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location 812
      }
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 824
      }
    )
    :location 822
  }
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location 863
      }
      {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location 875
      }
    )
    :location 873
  }
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 2
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 4
        :location 914
      }
      {VAR
        :varno 3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location 928
      }
    )
    :location 926
  }
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 3
        :location 968
      }
      {VAR

```

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:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 982
}
)
:location 980
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 1029
}
){CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1045
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 1043
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2146
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 599
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 595
}
:resno 1
:resname min
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
)
}

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [100 0 0 0 0 0 0 0 0 0 0]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b 1) :outer_rels (b 1) :right_rels (b 1) :orclause <> :rinfo_serial 1 :reval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :sublinkType 0 :testexpr <> :paramids <> :plan_id 3 :plan_name SubPlan\ 3 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 1) :args ({VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} :startupt_cost 0 :per_call_cost 121512.000000000003} {SUBPLAN :sublinkType 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {PARAM :paramkind 1 :paramid 4 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :paramids (i 4) :plan_id 6 :plan_name SubPlan\ 6 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe false :setParam <> :parParam <> :args <> :startupt_cost 121502.625000000001 :per_call_cost 0.0025})) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b 1) :outer_rels (b 1) :left_rels (b 1) :right_rels (b 1) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :p


```
aramcollid 0 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1)
:incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial 5 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
s (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 0 :paramid 0 :paramtype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0}
erator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0}
WARNING: max_parallel_hazard_walker: SubPlan plan_id=3 parallel_safe=0 > reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OPEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :CONST :consttype 23 :consttypmod -1 :constcollid 0 :consten 4 :constbyval true :constisnull false :location -1 :constval 6 [ 100 0 0 0 0 0 128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICINFO :clause {ALTERNATIVESUBPLAN :subplans {(SUBPLAN :sublinktype 0 :testexpr <> :paramids <> :plan_id 3 :plan_name SubPlan\ 3 :firstCollType 2278 :firstCollTypeMod -1 :firstCollCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 1) :args {(VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} :startup_cost 0 :per_call_cost 121512.00000000003} (SUBPLAN :sublinktype 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} (PARAM :paramkind 1 :paramid 4 :paramtype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} :paramids (i 4) :plan_id 6 :plan_name SubPlan\ 6 :firstCollType 23 :firstCollTypeMod -1 :firstCollCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe false :setParam <> :parParam <> :args <> :startup_cost 121502.750000 0001 :per_call_cost 0.0025}} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} (PARAM :paramkind 1 :paramid 0 :paramtype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} :paramids (i 8) :plan_id 6 :plan_name SubPlan\ 6 :firstCollType 23 :firstCollTypeMod -1 :firstCollCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe false :setParam <> :parParam <> :args <> :startup_cost 124.4875 :per_call_cost 0.0025}} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0}
WARNING: max_parallel_hazard_walker: SubPlan plan_id=7 parallel_safe=0 > reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 2 :varattno 6 :vartype 1 700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} :CONST :consttype 1700 :consttypmod -1 :constcollid 0 :consten 1 :constbyval false :constisnull false :location -1 :constval 6 [ 24 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICINFO :clause {ALTERNATIVESUBPLAN :subplans {(SUBPLAN :sublinktype 0 :testexpr <> :paramids <> :plan_id 7 :plan_name SubPlan\ 7 :firstCollType 2278 :firstCollTypeMod -1 :firstCollCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 7) :args {(VAR :varno 2 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625} (SUBPLAN :sublinktype 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 2 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 4 :location -1} (PARAM :paramkind 1 :paramid 8 :paramtype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 4 :location -1} :paramids (i 8) :plan_id 8 :plan_name SubPlan\ 8 :firstCollType 23 :firstCollTypeMod -1 :firstCollCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025}} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} | parallel_safe = false | reason = PROPAPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0}
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0} | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OPEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} :CONST :consttype 1042 :consttypmod -1 :constcollid 100 :consten 1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 4) :required_relds (b 4) :incompatible_relds (b) :outer_relds (b) :left_relds (b 4) :right_relds (b) :orclause <> :rinfo_serial 9 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :consten 1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0} :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0}
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OPEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} :CONST :consttype 1042 :consttypmod -1 :constcollid 100 :consten 1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 4) :required_relds (b 4) :incompatible_relds (b) :outer_relds (b 4) :left_relds (b 4) :right_relds (b) :orclause <> :rinfo_serial 9 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :consten 1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0} :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0}
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0} | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({AGGREF :aggfnoid 2146 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 1700 :aggargtypes (o 1700) :aggdirectargs <> :args {(TARGETENTRY :expr {VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}} :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :locatn -1})
WARNING: is_parallel_safe: OUTPUT qual = ({AGGREF :aggfnoid 2146 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 1700 :aggargtypes (o 1700) :aggdirectargs <> :args {(TARGETENTRY :expr {VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}} :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :locatn -1})
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 1700 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0}
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 1700 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 0} | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOns false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:colList <>
:rtable
{
(RANGE)BLENTRY
:alias
(ALIAS
:aliasname n
}
```

```

        :colnames <>
    }
}
:eref
{
    {ALIAS
        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{
    {ALIAS
        :aliasname r
        :colnames <>
    }
}
}
:eref
{
    {ALIAS
        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
    }
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{
    {FROMEXPR
        :fromlist (
            {RANGETBLREF
                :rtindex 1
            }
            {RANGETBLREF
                :rtindex 2
            }
        )
    }
}
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
    }
    :varnosyn 1
    :varattnosyn 1
    :location -1
    }
    {PARAM
        :paramkind 1
        :paramid 9
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
    }
}
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
    }
    {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
    }
}
:location -1
}
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
}
}

```

```
      :args (
        {VAR
          :varno 2
          :varattno 2
          :vartype 1042
          :vartypmod 29
          :varcollid 100
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 2
          :varattnosyn 2
          :location -1
        }
        {CONST
          :consttype 1042
          :consttypmod -1
          :constcollid 100
          :constlen -1
          :constbyval false
          :constisnull false
          :location -1
          :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
        }
      )
      :location -1
    }
  }
}

:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 1 :vartype 20 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=9 in_safe_list=0 > reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :retList <>
  :rtable (
    {RANGEtblENTRY
      :alias
      {ALIAS
        :aliasname n
        :colnames <>
      }
    }
    :eref
    {ALIAS
      :aliasname n
      :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
  )
  :rtekind 0
  :relid 17158
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablespace <>
  :lateral false
  :inFromCl true
  :securityQuals <>
}
{RANGEtblENTRY
  :alias
  {ALIAS
    :aliasname r
    :colnames <>
  }
}
:eref
{ALIAS
```

```

        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
    }
}
:rtakind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesamples <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>

```

```
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :co
nstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_
jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :cons
tvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :c
onstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause
<> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_
jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :con
stvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :va
rtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :
has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.00588223529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jd
omain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :
has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype
23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_
jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :var
nosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_b
ucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupCTE false
:isReturn false
:ctelist <>
:rttable (
{RANGEABLEENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtkind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGEABLEENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtkind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos (
{RTPERMISSIONINFO
:relid 17158
```

```

:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 12
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>

```

```

:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 12 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 12 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=12 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 12 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 12 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} )
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} ) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:costList <>
:table (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:reld 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:reld 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:teperminfos (
{RTEPERMISSIONINFO
:reld 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reld 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtdindex 1
}
{RANGETBLREF
:rtdindex 2
}
)
}

```

```

:quals (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
      }
      {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
      }
    )
    :location -1
  )
  {OPEXPR
    :opno 1054
    :opfuncid 1048
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 100
    :args (
      {VAR
        :varno 2
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
      }
      {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
      }
    )
    :location -1
  )
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
  }
  :resno 1
  :resname <>
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ABALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1}) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b 2) :right_relds (b 2) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :cons
tvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1}) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b 2) :left_relds (b 2) :right_relds (b) :orclause
<> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :con

```



```
stvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)}
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
)}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 3 :location -1)} (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 2 :varattnosyn 1 :location -1)} :location -1)} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 3 :location -1)} (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 2 :varattnosyn 1 :location -1)} :location -1)} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
)}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
)}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity true
  :hasGroupRTE false
  :isReturn false
  :cteList <>
  :table (
    {RANGELEFTENTRY
      :alias
      {ALIAS
        :aliasname s
        :colnames <>
      }
    }
  )
  :eref (
    {ALIAS
      :aliasname s
      :colnames ("s_supplek" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
    }
  )
  :rtekind 0
  :reld 17181
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals (
    {OEXPR
      :opno 1756
      :opfuncid 1720
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        (VAR
          :varno 1
          :varattno 6
          :vartype 1700
          :vartypmod 983046
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 6
          :location -1
        )
      )
    }
    {CONST
      :consttype 1700
      :consttypmod -1
      :constcollid 0
      :constlen -1
      :constbyval false
      :constisnull false
      :location -1
      :constvalue 6 [ 24 0 0 0 0 -128 ]
    }
  )
  :location -1
  }
  {ALTERNATIVESUBPLAN
    :subplans (
      {SUBPLAN
        :sublinkType 0
        :testexpr <>
        :paramids <>
        :plan_id 12
        :plan_name SubPlan\ 12
        :firstColType 2278
        :firstColTypmod -1
        :firstColCollation 0
        :useHashTable false
        :unknownEqFalse true
        :parallel_safe false
        :setParam <>
        :parParam (i 12)
        :args (
          (VAR
            :varno 1
            :varattno 4
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 4
            :location -1
          )
        )
      }
    )
    :startup_cost 0
    :per_call_cost 24.2625
  }
  {SUBPLAN
    :sublinkType 2
```

```

: testexpr
{
  OEXPR
  : opno 96
  : opfuncid 65
  : opresulttype 16
  : opretset false
  : opcollid 0
  : inputcollid 0
  : args (
    {
      VAR
      : varno 1
      : varattno 4
      : vartype 23
      : vartypmod -1
      : varcollid 0
      : varnullingrels (b)
      : varlevelsup 0
      : varreturningtype 0
      : varnosyn 1
      : varattnosyn 4
      : location -1
    }
    {
      PARAM
      : paramkind 1
      : paramid 13
      : paramtype 23
      : paramtypmod -1
      : paramcollid 0
      : location -1
    }
  )
  : location -1
}
: paramids (i 13)
: plan_id 13
: plan_name SubPlan\ 13
: firstColType 23
: firstColTypmod -1
: firstColCollation 0
: useHashTable true
: unknownEqFalse true
: parallel_safe true
: setParam <>
: parParam <>
: args <>
: startup_cost 24.4875
: per_call_cost 0.0025
}
)
))
}
)
: rteperinfos (
{
  RTEPERMISSIONINFO
  : relid 17181
  : inh true
  : requiredPerms 2
  : checkAsUser 0
  : selectedCols (b 0 13)
  : insertedCols (b)
  : updatedCols (b)
}
)
: jointree
{
  FROMEXPR
  : fromlist (
    {
      RANGETBLREF
      : rtindex 1
    }
  )
  : quals (
    {
      OEXPR
      : opno 96
      : opfuncid 65
      : opresulttype 16
      : opretset false
      : opcollid 0
      : inputcollid 0
      : args (
        {
          VAR
          : varno 1
          : varattno 1
          : vartype 23
          : vartypmod -1
          : varcollid 0
          : varnullingrels (b)
          : varlevelsup 0
          : varreturningtype 0
          : varnosyn 1
          : varattnosyn 1
          : location -1
        }
        {
          PARAM
          : paramkind 1
          : paramid 11
          : paramtype 23
          : paramtypmod -1
          : paramcollid 0
          : location -1
        }
      )
    }
    : location -1
  )
  {
    OEXPR
    : opno 1756
    : opfuncid 1720
    : opresulttype 16
    : opretset false
    : opcollid 0
    : inputcollid 0
    : args (
      {
        VAR
        : varno 1
        : varattno 6
        : vartype 1700
        : vartypmod 983046
        : varcollid 0
        : varnullingrels (b)
        : varlevelsup 0
        : varreturningtype 0
        : varnosyn 1
        : varattnosyn 6
        : location -1
      }
      {
        CONST
        : consttype 1700
        : consttypmod -1
        : constcollid 0
        : constlen -1
        : constbyval false
        : constisnull false
        : location -1
        : constvalue 6 [ 24 0 0 0 0 -128 ]
      }
    )
  }
  : location -1
}
)
}
: mergeActionList <>
: mergeTargetRelation 0
: mergeJoinCondition <>
: targetList <>
: override 0
: onConflict <>
: returningOldAlias <>
: returningNewAlias <>

```

```
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
```

```
WARNING: is_parallel_safe: INPUT qual = (({RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 6 :vartype 17 00 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbtyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :sublinkType 0 :testexpr <> :params <> :plan_id 12 :plan_name SubPlan} 12 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 12) :args {(VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625) {SUBPLAN :sublinkType 0 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbtyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 13 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :params (i 13) :plan_id 12 :plan_name SubPlan} 13 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 12) :args {(VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_domain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 11 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_domain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
```

```
WARNING: max_parallel_hazard_walker: SubPlan plan_id=12 parallel_safe=0 → reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = (({RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbtyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :sublinkType 0 :testexpr <> :params <> :plan_id 12 :plan_name SubPlan} 12 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 12) :args {(VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625) {SUBPLAN :sublinkType 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 13 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :params (i 13) :plan_id 12 :plan_name SubPlan} 13 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 12) :args {(VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbtyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_domain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 11 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_domain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = false | reason = PR
```

```
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
```

```
WARNING: Log 2: After pull up At starting of Path Generation:
```

```
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtkind 0
:reld 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtkind 0
:reld 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
}
:rtperminfos (
{RTEPERMISSIONINFO
:reld 17158
:inh true
:requiredPerms 2
:checkAsUser 0
}
```

```

:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 15
:paramtype 23
:paramstypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1040
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:varstype 1042
:varstypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:conststypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>

```

```
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 15 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 15 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=15 in_safe_list=0 => reason=outer_reference

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 15 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 15 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:costlist <>
:rttable {
:RANGETBLENTY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtckind 0
:reld 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
:RANGETBLENTY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtckind 0
:reld 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rtperminfos {
{RTEPERMISSIONINFO
:reld 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reld 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
}
:jointree
{FROMEXPR
:fromlist {
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
}
}
:quals {
{OEXPR
```

```

:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1}) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1}) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1})

```
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :va
rtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jd
omain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype
23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :var
nosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_b
ucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTF false
:isReturn false
:cteList <>
:table (
:RANGEBLENTTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17181
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ({
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 15
:plan_name SubPlan\ 15
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 15)
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:startup_cost 0
:per_call_cost 24.2625
}
{SUBPLAN
:subLinkType 2
:testexpr
{OEXPR
:opno 96
```

```

:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 16
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramids (i 16)
:plan_id 16
:plan_name SubPlan\ 16
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 24.4875
:per_call_cost 0.0025
}
)
})
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17181
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
)
:quals (
  {OPEXPR
    :opno 1756
    :opfuncid 1720
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
      }
      {CONST
        :consttype 1700
        :consttypmod -1
        :constcollid 0
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 6 [ 24 0 0 0 0 -128 ]
      }
    )
  }
  :location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>

```



```
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRUCTINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 17
00 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :const
byval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial
1 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRUCTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :sublinktype 0 :testexpr <> :paramids <> :plan_id 15
:plan_name SubPlan\ 15 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 15) :args ({VAR :varno 1 :varat
tno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625) {SUBPLAN :sublinkType
e 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varle
velsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 16) :plan_id
16 :plan_name SubPlan\ 16 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :p
er_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1)
:required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_sel
ec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hash
eqoperator 0} {RESTRUCTINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcol
lid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull fa
lse :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security
_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :e
val_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right
_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: SubPlan plan_id=15 parallel_safe=0 → reason=correlated_subplan

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRUCTINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1
700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :con
stbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fal
se :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_seri
al 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsi
ze -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRUCTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :sublinktype 0 :testexpr <> :paramids <> :plan_id 1
5 :plan_name SubPlan\ 15 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 15) :args ({VAR :varno 1 :vara
tno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625) {SUBPLAN :sublinkType
pe 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varl
evelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 16) :plan_id
16 :plan_name SubPlan\ 16 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :p
er_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b
1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_se
lec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_has
heqoperator 0} {RESTRUCTINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varco
llid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull f
alse :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security
_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :
eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :righ
t_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED

WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)

WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: (
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSublinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity true
  :hasGroupRTe false
  :isReturn false
  :isList <>
  :table (
    {RANGEBENTRY
    :alias <>
    :eref
    {ALIAS
    :aliasname part
    :colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
    "p_container" "p_retailprice" "p_comment")
    }
    :rtekind 0
    :reld 17169
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    {RANGEBENTRY
    :alias <>
    :eref
    {ALIAS
    :aliasname supplier
    :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
    s_acctbal" "s_comment")
    }
    :rtekind 0
    :reld 17181
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals ((
      {OPEXPR
      :opno 1756
      :opfuncid 1720
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
        :varno 2
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varattnosyn 2
        :varattnosyn 6
        :location -1
        }
        {CONST
        :consttype 1700
        :consttypmod -1
        :constcollid 0
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 6 [ 24 0 0 0 0 -128 ]
        }
        )
      }
      :location -1
    )
    {ALTERNATIVESUBPLAN
    :subplans (
      {SUBPLAN
      :sublinktype 0
      :testexpr <>
    )
  )
}
```

```

:paramIds <>
:plan_id 10
:plan_name SubPlan\ 10
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 9)
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location -1
  }
)
:startup_cost 0
:per_call_cost 24.2625
}
(SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 10
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramIds (i 10)
:plan_id 11
:plan_name SubPlan\ 11
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 24.4875
:per_call_cost 0.0025
}
)
}
(RANGEABLEENTRY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
)
:rtekind 0
:relid 17191
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
  {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 3
        :location -1
      }
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 100 0 0 0 0 0 0 ]
      }
    )
  }
)
:location -1
}
(ALTERNATIVESUBPLAN
:subplans (
  {SUBPLAN
    :subLinkType 0
    :testexpr <>
    :paramIds <>
    :plan_id 14
    :plan_name SubPlan\ 14
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse true

```

```

:parallel_safe false
:setParam <>
:parParam (i 11)
:args (
  {VAR
    :varno 3
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 2
    :location -1
  }
)
:startup_cost 0
:per_call_cost 121512.00000000003
}
{SUBPLAN
:sublinkType 2
:testexpr
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 2
        :location -1
      }
      {PARAM
        :paramkind 1
        :paramid 14
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
    :location -1
  }
:paramids (i 14)
:plan_id 17
:plan_name SubPlan\ 17
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam <>
:args <>
:startup_cost 121502.62500000001
:per_call_cost 0.0025
}
)
}
))
}
{RANGETBLENTY
:alias <>
:eref
  {ALIAS
    :aliasname nation
    :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
  }
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
  {ALIAS
    :aliasname region
    :colnames ("r_regionkey" "r_name" "r_comment")
  }
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17169
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 10 12 13)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17181
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17191
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 11)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17158
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10)
    :insertedCols (b)
    :updatedCols (b)
  }
)

```

```

}
{RTEPERMISSIONINFO
:reld 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 9)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
}
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 295
}
)
:location 293
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 318
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 330
}
)
:location 328
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 353
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 362
:constvalue 4 [ 15 0 0 0 0 0 0 0 ]
}
)
:location 360
}
{OEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE

```

```

:arg
{VAR
:varno 1
:varattno 5
:vartype 1043
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 377
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 389
:constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
}
)
:location 384
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 410
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 424
}
}
)
:location 422
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 448
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 462
}
}
)
:location 460
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 5
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 493
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 509
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
)
:location 507
}
{OPEXPR
:opno 1752
:opfuncid 1718
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0

```

```

:args (
  {VAR
    :varno 3
    :varattno 4
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 4
    :location 530
  }
  {SUBPLAN
    :sublinkType 4
    :testexpr <>
    :paramids <>
    :plan_id 9
    :plan_name SubPlan\ 9
    :firstColType 1700
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse false
    :parallel_safe false
    :setParam <>
    :parParam (i 0)
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 812
      }
    )
    :startup_cost 0
    :per_call_cost 48604936345.600006
  }
)
:location 544
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 2
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 6
        :location 51
      }
    }
  :resno 1
  :resname s_acctbal
  :ressortgroupref 1
  :resorigtbl 17181
  :resorigcol 6
  :resjunk false
  }
  {TARGETENTRY
    :expr
      {VAR
        :varno 2
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location 70
      }
    }
  :resno 2
  :resname s_name
  :ressortgroupref 3
  :resorigtbl 17181
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
    :expr
      {VAR
        :varno 4
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location 86
      }
    }
  :resno 3
  :resname n_name
  :ressortgroupref 2
  :resorigtbl 17158
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 102
      }
    }
  :resno 4
  :resname p_partkey
  :ressortgroupref 4
  :resorigtbl 17169
  :resorigcol 1
  :resjunk false
  }
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 3

```

```

:varstype 1042
:varystmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 17169
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:varstype 1043
:varystmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:varstype 1042
:varystmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:varstype 1043
:varystmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause {
{SORTGROUPCLAUSE
:t1SortGroupRef 1
:eqop 1752
:sortop 1756
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:t1SortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:t1SortGroupRef 3
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:t1SortGroupRef 4
:eqop 96
:sortop 97
:reverse_sort false
:nulls_first false
:hashable true
}
}
:limitOffset <>
:limitCount
{CONST
:consttype 20
:constystmod -1
:constcollid 0
:constlen 8
:constbval true
:constisnull false
:location -1
:constvalue 8 [ 100 0 0 0 0 0 0 0 ]
}
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0

```

```
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1209 :ofuncid 850 :presulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((RELABELTYPE :arg {VAR :varno 1 :vartype attno 5 :vartype 1043 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resulttype 25 :resulttupledim 1 :resultcollid 100 :relabelformat 2 :location -1}) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen 1 :constbyval true :constnullin false :location 1 :constvalue [ 0 4 0 0 0 37 66 82 65 83 83 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :right_relds (b 2) :orclause < :rinfo_serial 7 :eval_cost.startup 1 :eval_cost.per_tupl 0 :norm_select 1 :outer_select 1 :mergoeffamilies < :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0} (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args (VAR :varno 1 :varrattno 6 :vartype 23 :vartypmod 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constnullin false :location 1 :constvalue 4 [ 15 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_rels 1 :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :right_relds (b 2) :orclause < :rinfo_serial 8 :eval_cost.startup 1 :eval_cost.per_tupl 0 :norm_select 1 :outer_select 1 :mergoeffamilies < :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0} (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args (VAR :varno 1 :varrattno 6 :vartype 23 :vartypmod 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5))) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0} | parallel_safe = true | reason = PROPAGALATE_SAFE  
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1209 :ofuncid 850 :presulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((RELABELTYPE :arg {VAR :varno 1 :varrattno 5 :vartype 1043 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resulttype 25 :resulttupledim 1 :resultcollid 100 :relabelformat 2 :location -1}) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen 1 :constbyval true :constnullin false :location 1 :constvalue [ 0 4 0 0 0 37 66 82 65 83 83 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :right_relds (b 2) :orclause < :rinfo_serial 7 :eval_cost.startup 1 :eval_cost.per_tupl 0 :norm_select 1 :outer_select 1 :mergoeffamilies < :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0} (RESTRICINFO :clause (OPEXPR :opno 96 :ofuncid 65 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args (VAR :varno 1 :varrattno 6 :vartype 23 :vartypmod 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5))) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4 5))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0} | parallel_safe = true | reason = PROPAGALATE_SAFE  
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1756 :ofuncid 1720 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args (VAR :varno 2 :varrattno 6 :vartype 1700 :vartypmod 983946 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1}) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constnullin false :location 1 :constvalue 6 [ 2 4 0 0 0 -128 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :right_relds (b 2) :orclause < :rinfo_serial 1 :eval_cost.startup 1 :eval_cost.per_tupl 0 :norm_select 1 :outer_select 1 :mergoeffamilies < :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0} (RESTRICTINFO :clause (ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinktype 0 :testexpr <> :paramsid <> :plan_id 10 :plan_name SubPlan10 :firstCollType 2278 :firstCollTypmod -1 :firstCollCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParameter <> :parParam (i 9) :args (VAR :varno 2 :varrattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 4 :location -1})) :startup_cost 0 :per_call_cost 24.2625) (SUBPLAN :sublinktype 2 :testexpr (OPEXPR :opno 96 :ofuncid 65 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args (VAR :varno 2 :varrattno 4 :vartype 23 :vartypmod 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1}) (PARAM :paramkind 1 :paramid 10 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1)) :location -1) :paramids (i 10) :plan_id 11 :plan_name SubPlan11 :firstCollType 23 :firstCollTypmod -1 :firstCollCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParameter <> :parParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025)) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :right_relds (b 2) :orclause < :rinfo_serial 2 :eval_cost.startup 1 :eval_cost.per_tupl 0 :norm_select 1 :outer_select 1 :mergoeffamilies < :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0}  
WARNING: is_parallel_hazard_walker: SubPlan plan_id=10 parallel_safe=> reason=correlated_subplan  
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1756 :ofuncid 1720 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args (VAR :varno 2 :varrattno 6 :vartype 1700 :vartypmod 983946 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1}) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constnullin false :location 1 :constvalue 6 [ 2 4 0 0 0 -128 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :right_relds (b 2) :orclause < :rinfo_serial 1 :eval_cost.startup 1 :eval_cost.per_tupl 0 :norm_select 1 :outer_select 1 :mergoeffamilies < :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0} (RESTRICTINFO :clause (ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinktype 0 :testexpr <> :paramsid <> :plan_id 11 :plan_name SubPlan11 :firstCollType 23 :firstCollTypmod -1 :firstCollCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParameter <> :parParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025)) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :right_relds (b 2) :orclause < :rinfo_serial 2 :eval_cost.startup 1 :eval_cost.per_tupl 0 :norm_select 1 :outer_select 1 :mergoeffamilies < :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0} | parallel_safe = false | reason = PROPAGALATE_RESTRICTED  
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 521 :ofuncid 147 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args (VAR :varno 3 :varattnnos 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1}) (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constnullin false :location 1 :constvalue 4 [ 100 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b 3) :outer_relds (b 3) :right_relds (b 3) :orclause
```



```
(VAR :varno 0 :vartatno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 4 :varattosyn 1 :location -1}) | parallel_safe = true | reason = PROPAGALLES_SAFE
```

```
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :vartatno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 6 :locati  
on -1} (VAR :varno 2 :vartatno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattotno 2 :location -1} (VAR :varno 4 :vartatno 2 :va  
rtpe 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattotno 2 :location -1} (VAR :varno 1 :vartatno 1 :vartype 23 :vartypmod -1 :varcollid 0 :v  
arnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varattosyn 1 :location -1} (VAR :varno 1 :vartatno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :var  
returningtype 0 :varnosyn 1 :varattosyn 3 :location -1} (VAR :varno 2 :vartatno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattot  
syn 3 :location -1} (VAR :varno 2 :vartatno 5 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 5 :location -1} (VAR :varno 2 :va  
rattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 7 :location -1}))  
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :vartatno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 6 :locati  
ion -1} (VAR :varno 2 :vartatno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattotno 2 :location -1} (VAR :varno 4 :vartatno 2 :va  
rtpe 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattotno 2 :location -1} (VAR :varno 1 :vartatno 1 :vartype 23 :vartypmod -1 :varcollid 0 :  
varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varattosyn 1 :location -1} (VAR :varno 1 :vartatno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :va  
rreturningtype 0 :varnosyn 1 :varattosyn 3 :location -1} (VAR :varno 2 :vartatno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattot  
syn 3 :location -1} (VAR :varno 2 :vartatno 5 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 5 :location -1} (VAR :varno 2 :va  
rattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 7 :location -1}))  
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :vartatno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 6 :locati  
on -1} (VAR :varno 2 :vartatno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattotno 2 :location -1} (VAR :varno 4 :vartatno 2 :va  
rtpe 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattotno 2 :location -1} (VAR :varno 1 :vartatno 1 :vartype 23 :vartypmod -1 :varcollid 0 :v  
arnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varattosyn 1 :location -1} (VAR :varno 1 :vartatno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :va  
rreturningtype 0 :varnosyn 1 :varattosyn 3 :location -1} (VAR :varno 2 :vartatno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattot  
syn 3 :location -1} (VAR :varno 2 :vartatno 5 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 5 :location -1} (VAR :varno 2 :va  
rattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 2 :varattosyn 7 :location -1})) | parallel_safe = true | reason = PROPAGALLES_SAFE  
QUERY PLAN
```

```
Limit (cost=7120623197993.51..7120623197993.52 rows=1 width=271) (actual time=34676.819..34676.844 rows=100.00 loops=1)  
Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment  
Buffers: shared hit=6351847  
-> Sort (cost=7120623197993.51..7120623197993.52 rows=1 width=271) (actual time=34676.819..34676.836 rows=100.00 loops=1)  
Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment  
Sort Key: supplier.s_acctbal DESC, nation.n_name, supplier.s_name, part.p_partkey  
Sort Method: quicksort Memory: 77KB  
Buffers: shared hit=6351847  
-> Hash Join (cost=3586.99..7120623197993.50 rows=1 width=271) (actual time=13664.210..34676.396 rows=189.00 loops=1)  
Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment  
Hash Cond: ((partsupp.ps_supplycost = (SubPlan 9)) AND (partsupp.ps_partkey = part.p_partkey))  
Buffers: shared hit=6351838  
-> Nested Loop (cost=0.00..48604956119.52 rows=291 width=251) (actual time=5.126..7935.083 rows=70201.00 loops=1)  
Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, partsupp.ps_partkey, partsupp.ps_supplycost, nation.n_name  
Join Filter: (supplier.s_supkey = partsupp.ps_supkey)  
Rows Removed by Join Filter: 6219886  
Buffers: shared hit=9470  
-> Seq Scan on public.partsupp (cost=0.00..48604956119.52 rows=70201.00 loops=1) (actual time=1.441..385.514 rows=70201.00 loops=1)  
Output: partsupp.ps_partkey, partsupp.ps_supkey, partsupp.ps_avlity, partsupp.ps_supplycost, partsupp.ps_comment  
Filter: ((partsupp.ps_avlity > 180) AND (ANY (partsupp.ps_supkey = (hashed SubPlan 17).col1)))  
Rows Removed by Filter: 329799  
Buffers: shared hit=8906  
SubPlan 17  
-> Seq Scan on public.supplier s_1 (cost=0.00..121499.50 rows=1250 width=4) (actual time=0.049..1.304 rows=887.00 loops=1)  
Output: s_1.s_nationkey  
Filter: (((s_1.s_acctbal * '0')::numeric) AND (ANY (s_1.s_nationkey = (hashed SubPlan 16).col1)) AND (s_1.s_acctbal > '0')::numeric))  
Rows Removed by Filter: 4113  
Buffers: shared hit=114  
SubPlan 16  
-> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.028..0.032 rows=5.00 loops=1)  
Output: n_3.n_nationkey  
Hash Cond: (n_3.n_regionkey = r_3.r_regionkey)  
Buffers: shared hit=2  
-> Seq Scan on public.nation n_3 (cost=0.00..11.70 rows=170 width=8) (actual time=0.007..0.009 rows=25.00 loops=1)  
Output: n_3.n_nationkey, n_3.n_name, n_3.n_regionkey, n_3.n_comment  
Filters: shared hit=1  
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.014..0.014 rows=1.00 loops=1)  
Output: r_3.r_regionkey  
Buckets: 1024 Batches: 1 Memory Usage: 9KB  
Filters: shared hit=1  
-> Seq Scan on public.region r_3 (cost=0.00..12.12 rows=1 width=4) (actual time=0.008..0.009 rows=1.00 loops=1)  
Output: r_3.r_regionkey  
Filter: (r_3.r_name = 'EUROPE')::bpchar)  
Rows Removed by Filter: 4  
Filters: shared hit=1  
-> Materialize (cost=0.00..121528.61 rows=7 width=245) (actual time=0.000..0.031 rows=887.00 loops=70201)  
Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supkey, nation.n_name  
Storage: Memory Maximum Storage: 196KB  
Filters: shared hit=564  
-> Nested Loop (cost=0.00..121528.58 rows=7 width=245) (actual time=5.052..5.978 rows=887.00 loops=1)  
Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supkey, nation.n_name  
Join Filter: (supplier.s_nationkey = nation.n_nationkey)  
Rows Removed by Join Filter: 3548  
Filters: shared hit=564  
-> Nested Loop (cost=0.00..25.95 rows=1 width=108) (actual time=0.009..0.030 rows=5.00 loops=1)  
Output: nation.n_name, nation.n_nationkey  
Join Filter: (nation.n_regionkey = region.r_regionkey)  
Rows Removed by Join Filter: 20  
Filters: shared hit=2  
-> Seq Scan on public.region (cost=0.00..12.12 rows=1 width=4) (actual time=0.003..0.010 rows=1.00 loops=1)  
Output: region.r_regionkey, region.r_name, region.r_comment  
Filter: (region.r_name = 'EUROPE')::bpchar)  
Rows Removed by Filter: 4  
Filters: shared hit=1  
-> Seq Scan on public.nation (cost=0.00..11.70 rows=170 width=112) (actual time=0.004..0.014 rows=25.00 loops=1)  
Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment  
Filters: shared hit=1  
-> Seq Scan on public.supplier (cost=0.00..121487.00 rows=1250 width=145) (actual time=0.008..1.129 rows=887.00 loops=5)  
Output: supplier.s_supkey, supplier.s_name, supplier.s_address, supplier.s_nationkey, supplier.s_phone, supplier
```

```

Rows Removed by Join Filter: 632
Buffers: shared hit=6339260
-> Seq Scan on public.partsupp partsupp_1 (cost=0.00..48604814792.00 rows=2 width=10) (actual time=26.681..36.987 rows=0.71 loops=721)
  Output: partsupp_1.ps_partkey, partsupp_1.ps_supkey, partsupp_1.ps_availqty, partsupp_1.ps_supplycost, partsupp_1.ps_comment
  Filter: ((partsupp_1.ps_availqty > 100) AND (part_p_partkey = partsupp_1.ps_partkey) AND (ANY (partsupp_1.ps_supkey = (hashed SubPlan 6).col1)))
  Rows Removed by Filter: 399999
  Buffers: shared hit=6339146
  SubPlan 6
    -> Seq Scan on public.supplier s (cost=0.00..121499.50 rows=1250 width=4) (actual time=0.039..1.293 rows=887.00 loops=1)
      Output: s.s_supkey
      Filter: ((s.s_acctbal > '0'::numeric) AND (ANY (s.s_nationkey = (hashed SubPlan 5).col1)) AND (s.s_acctbal > '0'::numeric))
      Rows Removed by Filter: 4113
      Buffers: shared hit=114
      SubPlan 5
        -> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.025..0.030 rows=5.00 loops=1)
          Output: n.n_nationkey
          Hash Cond: (n.n_regionkey = r.r_regionkey)
          Buffers: shared hit=2
          -> Seq Scan on public.nation n (cost=0.00..11.70 rows=170 width=8) (actual time=0.006..0.008 rows=25.00 loops=1)
            Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment
            Buffers: shared hit=1
          -> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.010..0.011 rows=1.00 loops=1)
            Output: r.r_regionkey
            Buckets: 1024 Batches: 1 Memory Usage: 9kB
            Buffers: shared hit=1
            -> Seq Scan on public.region r (cost=0.00..12.12 rows=1 width=4) (actual time=0.004..0.005 rows=1.00 loops=1)
              Output: r.r_regionkey
              Filter: (r.r_name = 'EUROPE'::bpchar)
              Rows Removed by Filter: 4
              Buffers: shared hit=1
-> Materialize (cost=0.00..121493.25 rows=1250 width=8) (actual time=0.000..0.034 rows=887.00 loops=514)
  Output: supplier_1.s_supkey, supplier_1.s_nationkey
  Storage: Memory Maximum Storage: 58kB
  Buffers: shared hit=114
-> Seq Scan on public.supplier supplier_1 (cost=0.00..121487.00 rows=1250 width=8) (actual time=0.077..1.259 rows=887.00 loops=1)
  Output: supplier_1.s_supkey, supplier_1.s_nationkey
  Filter: ((supplier_1.s_acctbal > '0'::numeric) AND (ANY (supplier_1.s_nationkey = (hashed SubPlan 8).col1)))
  Rows Removed by Filter: 4113
  Buffers: shared hit=114
  SubPlan 8
    -> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.038..0.043 rows=5.00 loops=1)
      Output: n_1.n_nationkey
      Hash Cond: (n_1.n_regionkey = r_1.r_regionkey)
      Buffers: shared hit=2
      -> Seq Scan on public.nation n_1 (cost=0.00..11.70 rows=170 width=8) (actual time=0.006..0.007 rows=25.00 loops=1)
        Output: n_1.n_nationkey, n_1.n_name, n_1.n_regionkey, n_1.n_comment
        Buffers: shared hit=1
      -> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.019..0.021 rows=1.00 loops=1)
        Output: r_1.r_regionkey
        Buckets: 1024 Batches: 1 Memory Usage: 9kB
        Buffers: shared hit=1
        -> Seq Scan on public.region r_1 (cost=0.00..12.12 rows=1 width=4) (actual time=0.007..0.008 rows=1.00 loops=1)
          Output: r_1.r_regionkey
          Filter: (r_1.r_name = 'EUROPE'::bpchar)
          Rows Removed by Filter: 4
          Buffers: shared hit=1
-> Seq Scan on public.nation nation_1 (cost=0.00..11.70 rows=170 width=8) (actual time=0.002..0.003 rows=25.00 loops=514)
  Output: nation_1.n_nationkey, nation_1.n_name, nation_1.n_regionkey, nation_1.n_comment
  Buffers: shared hit=514
-> Seq Scan on public.region region_1 (cost=0.00..12.12 rows=1 width=4) (actual time=0.001..0.002 rows=1.00 loops=514)
  Output: region_1.r_regionkey, region_1.r_name, region_1.r_comment
  Filter: (region_1.r_name = 'EUROPE'::bpchar)
  Rows Removed by Filter: 4
  Buffers: shared hit=514

```

```

Planning:
  Buffers: shared hit=228
Planning Time: 35.119 ms
Execution Time: 34677.190 ms
(182 rows)

```

=====

BENCHMARK SUMMARY

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```

Runs          : 10
Average Planning : 31.538 ms
Average Execution : 34645.525 ms

```

```

Generated Files:
  run10_explain.txt
  summary.txt
adityadave@Adityas-MacBook-Air-3 Q2&Q21 %

```